

PHIL 342: Introduction to Groups and Choices

Brian Weatherson

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Key Points

- This course examines philosophical questions that arise when individual choices interact.
- The course has two main parts: pooling (how we form collective attitudes from individual attitudes) and modeling (what simple models reveal about social structures).
- We'll study voting methods, preference aggregation, opinion pooling, game theory, signaling, and models of scientific communities.
- Three textbooks are available electronically through the library: *Opinion Pooling* by Lee Elkin and Richard Pettigrew, *Modeling Scientific Communities* by Cailin O'Connor, and *Essentials of Game Theory* by Kevin Leyton-Brown and Yoav Shoham.
- Assessment consists of four short assignments (10% each), two essays (25% each), and discussion section participation (10%)

Course Description

This course introduces topics at the intersection of philosophy, game theory, and social choice theory. We examine how individual choices and judgments combine to form group decisions, and what mathematical models can teach us about social phenomena.

The first half focuses on pooling problems: when we try to aggregate individual beliefs or preferences into collective judgments, we encounter surprising paradoxes and impossibility results. We'll

explore voting methods, Arrow's theorem, judgment aggregation, and methods for pooling probabilistic opinions.

The second half looks at the use mathematical and computational models to understand social dynamics. The topics we'll cover include focal points, signaling in markets, iterated games, and agent-based models of scientific communities and knowledge production.

Required Materials

There are three textbooks for the course. All of them are available electronically through the library (and the first is open access, so doesn't even need a library login).

- Lee Elkin and Richard Pettigrew, [Opinion Pooling](#)
- Cailin O'Connor, [Modeling Scientific Communities](#) (if you are off-campus you can access this via [the library](#))
- Kevin Leyton-Brown and Yoav Shoham, [Essentials of Game Theory](#)

Course Requirements

- There will be 4 short assignments, each worth 10% of the grade
- There will be 2 essays, each worth 25% of the grade.
- Lecture participation will not be graded, and we won't take attendance here. But 10% of the grade is for discussion section participation (and any other assignments the discussion section leader chooses to include).

Summary of Grading System

1. Short assignments - 4 assignments, 10% each: 40%
2. Essays - 2 essays, 25% each: 50%
3. Discussion section participation. (10%)

Canvas

There is a Canvas site for this course, which can be accessed from <https://canvas.umich.edu>. Course documents will be available from this site. Please make sure that you can access this site. Consult the site regularly for announcements, including changes to the course schedule. And there are many tools on the site to communicate with each other, and with me.

There is also a github site for the course which contains more documents, and it is at <https://bweatherson.github.io/W26-Phil342-site/>.

Class Schedule

Readings should be done before the scheduled class.

Week 1: Introduction

Wednesday, January 07

Topic: Introduction

Reading: None.

NOTE: No discussion section this week.

Week 2: Voting

Monday, January 12

Topic: Voting Methods

Reading: Eric Pacuit, [Voting Methods](#)

Wednesday, January 14

Topic: Arrow's Theorem

Reading: Michael Morreau, [Arrow's Theorem](#)

Week 3: Arrow's Theorem

Monday, January 19

No class for MLK Day

Wednesday, January 21

Topic: Responses to Arrow

Reading: No new reading

Week 4: Sen's Paradox

Monday, January 26

Topic: Sen's Paradox

Reading: Amartya Sen, [The Impossibility of a Paretian Liberal](#)

Suggested Reading: Amartya Sen, [Liberty and Social Choice](#)

Wednesday, January 28

No lecture, because I'm away at a conference. There will be a discussion section.

Assignment: First short assignment due Thursday, January 29 at 5pm.

Week 5: Judgment Aggregation

Monday, February 02

Topic: From Belief to Probability

Reading: *Opinion Pooling*, Ch. 1 and section 2.1.

Suggested Reading:

- David Christensen, [Epistemology of Disagreement: The Good News](#)
- Adam Elga, [Reflection and Disagreement](#)
- Thomas Kelly, [The Epistemic Significance of Disagreement](#)

Wednesday, February 04

Topic: Reasons for Aggregation

Reading: *Opinion Pooling*, Rest of chapter 2.

Suggested Reading:

- Roger White, [Epistemic Permissiveness](#)
- Miriam Schoenfield, [Permission to Believe: Why Permissivism Is True and What It Tells Us About Irrelevant Influences on Belief](#)

Week 6: Constraints on Aggregation

Monday, February 09

Topic: Motivating the Constraints

Reading: *Opinion Pooling*, Chapter 3.

Wednesday, February 11

Topic: Linear and Geometric Pooling

Reading: *Opinion Pooling*, section 4.1.

Suggested Reading:

- John Hawthorne, Lara Buchak, and Jeffrey Sanford Russell, [Groupthink](#)
- Jan-Willem Romeijn, [An Interpretation of Weights in Linear Opinion Pooling](#)
- Jean Baccelli & Rush T. Stewart, [Support for Geometric Pooling](#)

Week 7: Evaluating the Strategies

Monday, February 16

Topic: Multiplicative Pooling

Reading: *Opinion Pooling*, sections 4.2-4.3

Wednesday, February 18

Topic: Group Rationality

Reading: *Opinion Pooling*, chapter 5.

Suggested Reading:

- James Joyce, [A Nonpragmatic Vindication of Probabilism](#)
- Rush T. Stewart & Ignacio Ojea Quintana, [Learning and Pooling, Pooling and Learning](#)

Assignment: Second short assignment due Thursday, February 19 at 5pm.

Week 8: Schelling

Monday, February 23

Topic: Focal Points

Reading: Judith Mehta, Chris Starmer, and Robert Sugden, [The Nature of Salience: An Experimental Investigation of Pure Coordination Games](#)

Suggested Reading: Robert Sugden, [A Theory of Focal Points](#)

Wednesday, February 25

Topic: Segregation

Reading: Vi Hart and Nicky Case, [Parable of the Polygons](#)

Suggested Reading: Thomas Schelling, [Dynamic Models of Segregation](#)

Essay: First essay due Thursday, February 26 at 5pm.

Week 9: Game Theory

Monday, March 09

Topic: One-shot games

Reading: *Essentials of Game Theory*, Chapters 1 and 2

Wednesday, March 11

Topic: Dynamic games

Reading: *Essentials of Game Theory*, Chapters 4 and 5

Week 10: Axelrod

Monday, March 16

Topic: Iterated Prisoners' Dilemma

Reading: Robert Axelrod and William D. Hamilton, [The Evolution of Cooperation](#)

Suggested reading: Robert Axelrod, [Effective Choice in the Prisoner's Dilemma](#)

Wednesday, March 18

Topic: Modern Approaches to IPD

Reading: Alexander J. Stewart and Joshua B. Plotkin, [Extortion and cooperation in the Prisoner's Dilemma](#)

Suggested Reading: William H. Press and Freeman J. Dyson, [Iterated Prisoner's Dilemma contains strategies that dominate any evolutionary opponent](#)

Assignment: Third short assignment due Thursday, March 19 at 5pm.

Week 11: Signals

Monday, March 23

Topic: Signals in car markets

Reading: George Akerlof, [The Market for Lemons](#)

Wednesday, March 25

Topic: Signals in employment markets

Reading: Michael Spence, [Signaling in Retrospect and the Informational Structure of Markets](#)

Suggested Reading: Michael Spence, [Job Market Signaling](#)

Week 12: Social Networks and Scientific Knowledge

Monday, March 30

Topic: Models of evidence sharing

Reading:

- *Modeling Scientific Communities*, Chapter 4 through 4.2.1.
- Kevin J. S. Zollman, [The Communication Structure of Epistemic Communities](#)

Wednesday, April 01

Topic: Evaluating models of evidence sharing

Reading: *Modeling Scientific Communities*, rest of Chapter 4.

Suggested Reading: Sarita Rosenstock, Justin Bruner, and Cailin O'Connor, [In Epistemic Networks, Is Less Really More?](#).

Week 13: The Credit Economy

Monday, April 06

Topic: The Matthew Effect

Reading:

- *Modeling Scientific Communities*, Chapter 2 through 2.6. (Section 2.6 is the most important part.)
- Robert K. Merton, [The Matthew Effect in Science](#)

Wednesday, April 08

Topic: Credit and Bias

Reading:

- *Modeling Scientific Communities*, rest of Chapter 2.
- Remco Heesen, [Academic Superstars: Competent or Lucky?](#)

Suggested Reading: Hannah Rubin, [Structural causes of citation gaps](#)

Assignment: Fourth short assignment due Thursday, April 09 at 5pm.

Week 14: Epistemic Landscapes

Monday, April 13

Topic: Landscapes and exploration

Reading:

- *Modeling Scientific Communities*, Chapter 5 through 5.1.
- Michael Weisberg and Ryan Muldoon, [Epistemic Landscapes and the Division of Cognitive Labor](#)

Wednesday, April 15

Topic: Rugged epistemic landscapes

Reading:

- *Modeling Scientific Communities*, rest of chapter 5.
- Johanna Thoma, [The Epistemic Division of Labor Revisited](#)

There will be a revision class on Monday, April 20, with no new material, to go over any questions people have about the final essay.

FINAL ESSAY: Due Tuesday, April 21 at 5pm.

Plagiarism

Although team-work, and even co-authorship, is encouraged, plagiarism is strictly prohibited. You are responsible for making sure that none of your work is plagiarized. Be sure to cite work that you use, both direct quotations and paraphrased ideas. Any citation method that is tolerably clear is permitted, but if you'd like a good description of a citation scheme that works well in philosophy, look at <http://bit.ly/VDhRJ4>.

You are encouraged to discuss the course material, including assignments, with your classmates, but all written work that you hand in under your own name must be your own. If work is handed in as the work of two people, you are affirming that each person did a fair share of the work. (Note that when you're submitting work on Canvas, you have to each submit the paper, even if it is co-authored. That way Canvas knows that everyone has turned in work.)

You should also be familiar with the academic integrity policies of the College of Literature, Science & the Arts at the University of Michigan, which are available here: <http://www.lsa.umich.edu/academicintegrity/>. Violations of these policies will be reported to the Office of the Assistant Dean for Student Academic Affairs, and sanctioned with a course grade of F.

Disability

The University of Michigan abides by the Americans with Disabilities Act of 1990, Section 504 of the Rehabilitation Act of 1973, and other applicable federal and state laws that prohibit discrimination on the basis of disability, which mandate that reasonable accommodations be provided for qualified students with disabilities.

If you have a disability, and may require some type of instructional and/or examination accommodation, please contact me early in the semester. If you have not already done so, you will also need to register with the Office of Services for Students with Disabilities. The office is located at G664 Haven Hall.

For more information on disability services at the University of Michigan, go to <http://ssd.umich.edu>.